Tendon-driven anthropomorphic hand with Bowden-tube tendon routing inside the hand skeleton, two-part antagonistic spools, direct finger coupling on the spool and a removable thumb-rotation actuator platform, for robotic and prosthetic use

- **Author:** Vittorio Lumare (vittorio.lumare@robotgarage.org)

- **Project name:** Yeah Hand (Rebelia V2)
- **First public disclosure of V2:** 26 November 2025, https://github.com/opsobot/rebelia/tree/V2 ; repository https://github.com/opsobot/yeah-hand and release v1.2 on 14 September 2026; assembly instructions Rev 1.8 on 5 October 2026
- **Previous version:** Rebelia V1, OSHWA IT000021, https://doi.org/10.5281/zenodo.23394100 , technical disclosure published on Technical Disclosure Commons

- **License of this document:** CC BY 4.0

Abstract

An anthropomorphic hand, manufacturable by 3D printing, usable as a robotic end effector and as an active prosthetic hand. Five rotary actuators are housed inside an internal skeleton of the hand. Each finger is driven by one actuator through an antagonistic pair of tendon ends, one for flexion and one for extension. Each tendon end runs inside its own low-friction Bowden tube (PTFE) from the finger to the actuator; the tubes are seated in holes, pass-through tunnels and slots formed in the printed skeleton, which separate and guide the tendons without a separate tendon barrier. On each actuator flange a two-part spool, made of a flexion spool and an extension spool stacked and fixed by the same screws, takes up one tendon end while releasing the other. The ring and little fingers share one actuator, with the tendon ends of both fingers fixed directly to the same spool. The thumb-rotation actuator, carrying the thumb base on its flange, is mounted on a separate platform inserted into the skeleton. Fingers are modular, with tendons pre-installed. The controller reads position, load, voltage, temperature and current from each actuator, performs whole-hand calibration and force-limited grasps, streams feedback for ROS, and supports two-channel EMG control in the prosthetic version.

Keywords

robotic hand; prosthetic hand; artificial hand; gripper; end effector; tendon-driven; cable-driven; Bowden cable; Bowden tube; PTFE tube; sheath; antagonistic tendons; flexor tendon; extensor tendon; spool; two-part spool; stepped spool; capstan; underactuated; finger coupling; thumb opposition; actuator platform; modular finger; identical fingers; interchangeable finger; servo load feedback; tendon calibration; force-limited grasp; ROS; EMG; myoelectric prosthesis; 3D-printed hand

Background

In tendon-driven hands, tendons that run freely inside the hand rub against each other and against the structure, change length when other parts move, and need barriers or pulleys to stay separated. Spools that carry both tendon ends in one piece are difficult to thread and tension during assembly. Coupling two fingers to one actuator usually needs extra parts (clamps, whiffletrees, sliding loops). Thumb actuators fixed directly in the main structure are hard to assemble and replace.

Description

1. Overall architecture (Figs. 1, 2)

The hand comprises an internal skeleton, a palm part, upper and lower covers, four upper fingers, a thumb and a robot or prosthetic interface. Five serial bus servomotors are housed in slots of the skeleton: index flexion, middle flexion, ring-and-little flexion, thumb flexion and thumb rotation. Left and right hand versions are mirrored. The robotic version is mounted on a robot through a flange according to ISO 9409-1-50-4-M6 (other flanges: ISO 9409-1-40, PCD-40); 12 V power comes from the robot interface and commands are wireless.

2. Thumb rotation servo platform (Fig. 3)

The thumb base is bolted to the output flange of the thumb rotation servo. The servo is screwed onto a separate platform (TRS platform), and the platform with servo and thumb base is inserted into the skeleton as a sub-assembly and fixed with screws. The thumb flexion tendon is driven by the thumb flexion servo in the skeleton.

3. Bowden-tube tendon routing (Fig. 4)

Each upper finger has a flexion end and an extension end of its tendon. Each end runs inside its own PTFE tube. For the flexion end, the tube enters an outlet hole on the rear of the skeleton, crosses to an inlet hole on the front near the finger slot, and is seated in an initial slot; its other end is seated in a final slot near the spool, passing where needed through pass-through tunnels formed in the skeleton. For the extension end, the tube runs from an initial slot on the rear of the skeleton, through tunnels, to a final slot near the spool. Tubes are cut to length. The skeleton geometry (holes, tunnels, slots) thus defines and fixes the path of every tendon end; tendons are separated from each other and from moving parts, and the path length is independent of the motion of other parts. A separate tendon barrier, used in V1, is no longer needed. Tendons are threaded through the tubes with a loop of nylon line used as a needle.

4. Two-part antagonistic spool (Figs. 5, 6)

Each flexion servo carries a spool assembly made of two parts mounted one above the other: a flexion spool (lower, with a stepped profile having its smaller diameter at the bottom) and an extension spool (upper). During installation the two parts are held together only by interlocking and by screws inserted but not tightened. The flexion end of the tendon is threaded through a hole of the flexion spool and out through the corresponding hole of the extension spool; the extension end is threaded through holes of the extension spool. With the servo in the tendon-installation position, the assembly is screwed to the

servo flange; the extension end is wound one turn clockwise around the extension spool, pulled until the finger is straight, wrapped around a bolt and locked by tightening the bolt; the flexion end is then pulled taut without moving the finger, wrapped around a second bolt and locked. Rotation of the servo in one direction winds the flexion end and unwinds the extension end (closing); the opposite rotation opens the finger. Both directions are actively driven, without return springs. The two parts can have different diameters so that take-up and release match the finger kinematics. The thumb uses a single spool carrying both ends.

5. Direct ring–little coupling on the spool

The ring and little fingers share one servo. The flexion ends of both fingers are fixed together to the flexion part of the same two-part spool, and the extension ends of both fingers to the extension part, without clamps or sliding loops (used in V1). The ring and little fingers therefore close and open together, driven by the same spool rotation.

6. Modular, identical upper fingers

The four upper fingers (index, middle, ring, little) use one and the same finger design: identical phalanges, joints, inserts, covers and pads, interchangeable between positions. One finger module is produced and stocked for all four positions, which reduces part count, tooling (including silicone pad moulds), assembly steps and spare parts. Each finger is supplied with its tendon already threaded through the phalanges and anchored at the distal phalanx. The finger is inserted into its slot on the skeleton and fixed with a screw and a nut; the free flexion and extension ends are then pulled into their Bowden tubes and fixed to the spool. Fingers can be replaced individually. Grip surfaces use rigid and flexible printed polymers with replaceable silicone pads.

7. Control and feedback (Figs. 7, 8)

The controller (ESP32) commands the servos over a serial bus and reads from each servo position, load, voltage, temperature and current. Functions of the robotic firmware (RebeliaRobot):

- whole-hand calibration: each servo, including thumb rotation, is driven towards each end of its travel until its load limit is reached, moved back by a small increment, and that position is stored as the range limit (compensating tendon length and stretch); per-servo ID registration programs;
- grasp presets (power, small power, hook/"monkey", pinch, relax) driven by a single 0–100% closure value, and RAW control of each degree of freedom;
- force-limited closing, each finger stopping when its load reaches a limit;
- protection thresholds for over-current and over-temperature;
- ROS mode: commands set the five degrees of freedom directly ("ROS a b c d e"), and the controller streams position, load, voltage, temperature and current of all servos about every 100 ms;
- tendon installation mode, moving servos to predefined positions;
- left/right hand selected in firmware; Bluetooth and Wi-Fi links;

Prosthetic firmware (RebeliaHuman, same as in Rebelia V1): two-channel EMG control, where the FFT intensity of each channel is compared with its own threshold to open, close or hold the hand, with grasp presets, force-limited closing, automatic relax and grasp

selection by manual thumb positioning. The automatic calibration routine belongs to the robotic firmware only.

Figures

Fig. 2 is rendered from the V2 STL files of release v1.2. Figs. 1 and 3–6 are taken from the V2 assembly instructions (Rev 1.8). Figs. 7–8 are drawn from the published firmware.

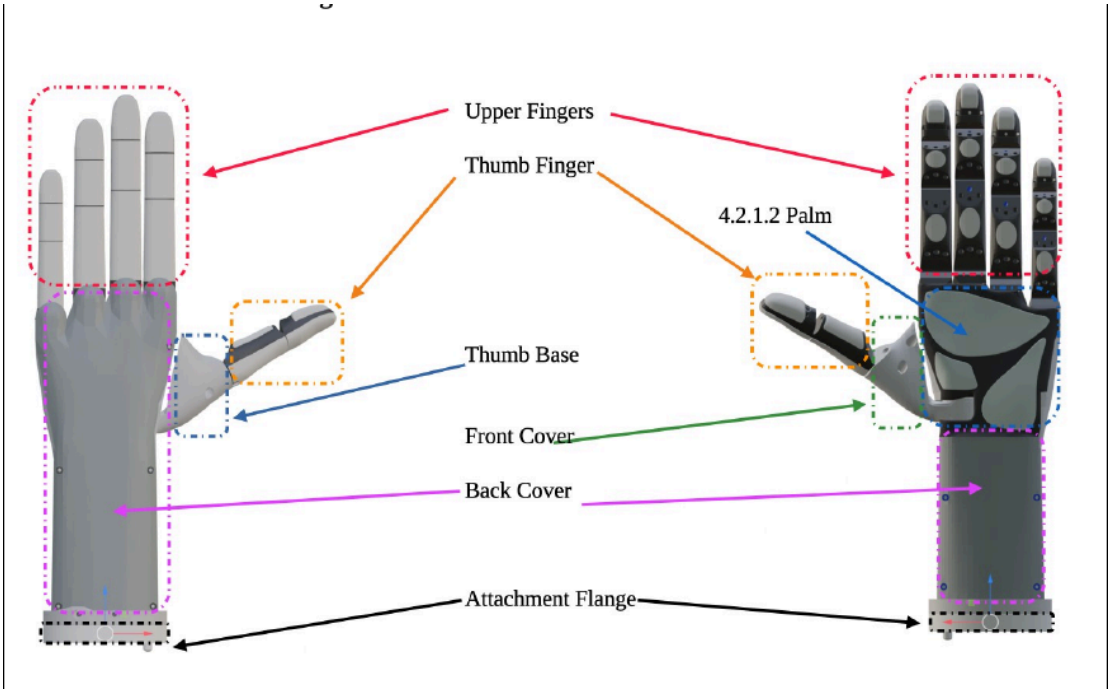


Fig. 1 — Overall view: upper fingers, thumb, thumb base, palm, covers and attachment flange.

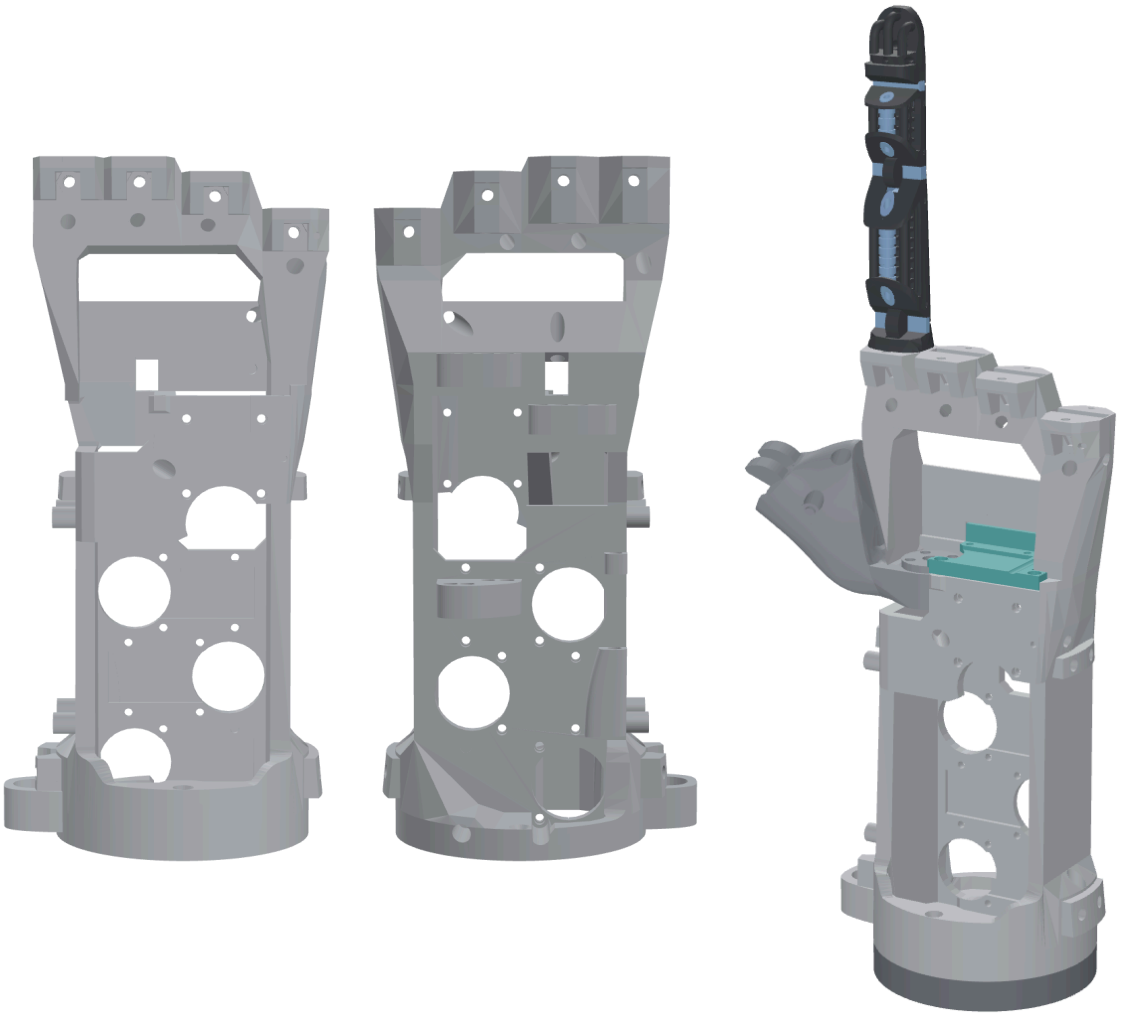


Fig. 2 — V2 skeleton, front and rear, with holes, tunnels and slots for the Bowden tubes (left); skeleton with TRS platform (green), thumb base, index finger and ISO 9409 flange (right). Rendered from the V2 STL files.



Fig. 3 — Thumb base on the thumb rotation servo; servo and thumb base on the TRS platform; TRS platform installed in the skeleton.

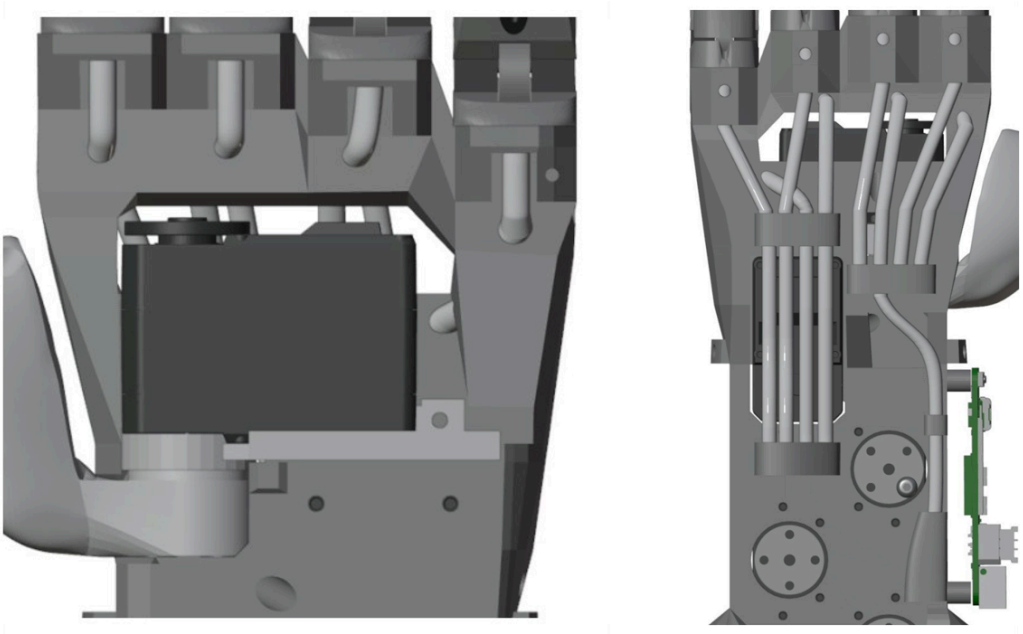


Fig. 4 — Bowden tubes installed in the skeleton, front and rear views.

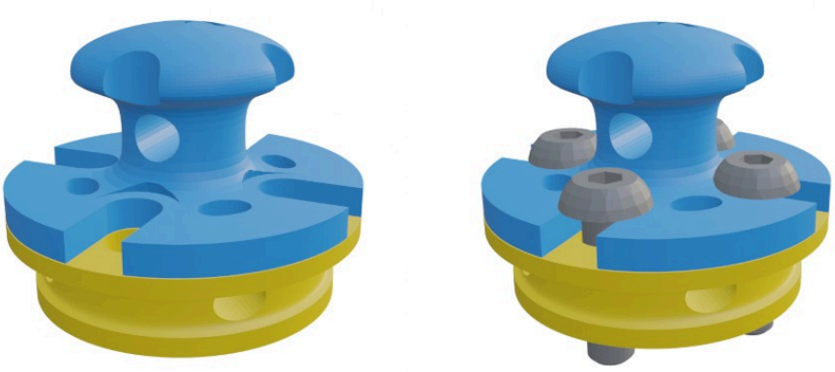


Fig. 5 — Two-part spool: extension spool (blue, top) and flexion spool (yellow, bottom), held together by the bolts (left) and assembled (right).

Figure 5.60: Spool assemblies mounted with arrow pointing up

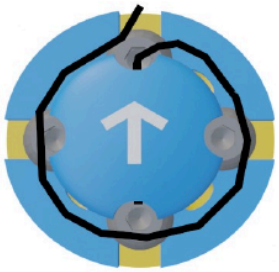


Figure 5.61: Pre-winding operation for the tendon's extension end

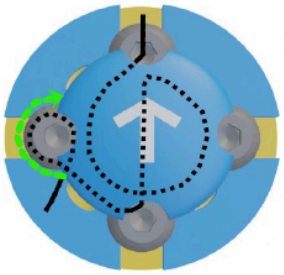


Figure 5.62: Extension end: Rotation around the bolt

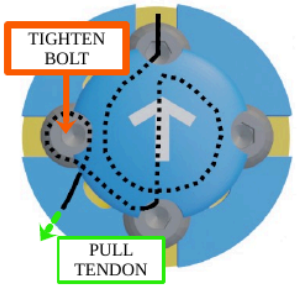


Figure 5.63: Extension end – Pulling and securing

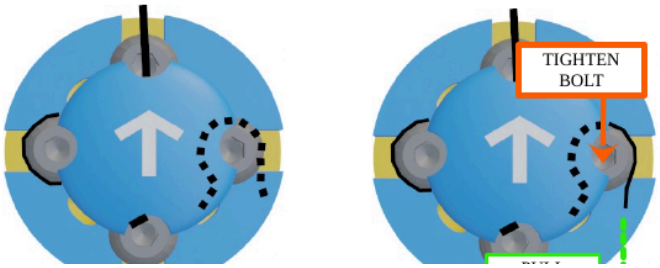


Fig. 6 — Tendon installation on the two-part spool: pre-winding, rotation around the bolt, pulling and locking of the extension end and of the flexion end.

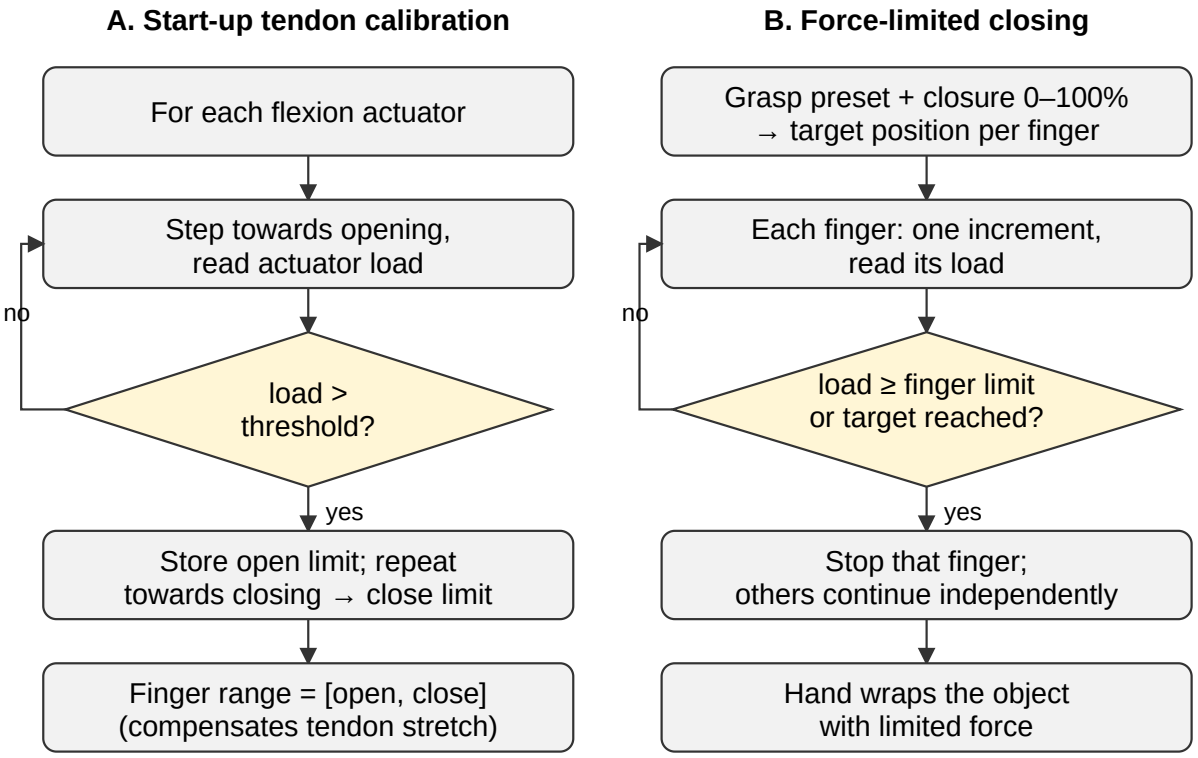


Fig. 7 — Flow chart of calibration (A) and force-limited closing (B), robotic firmware.

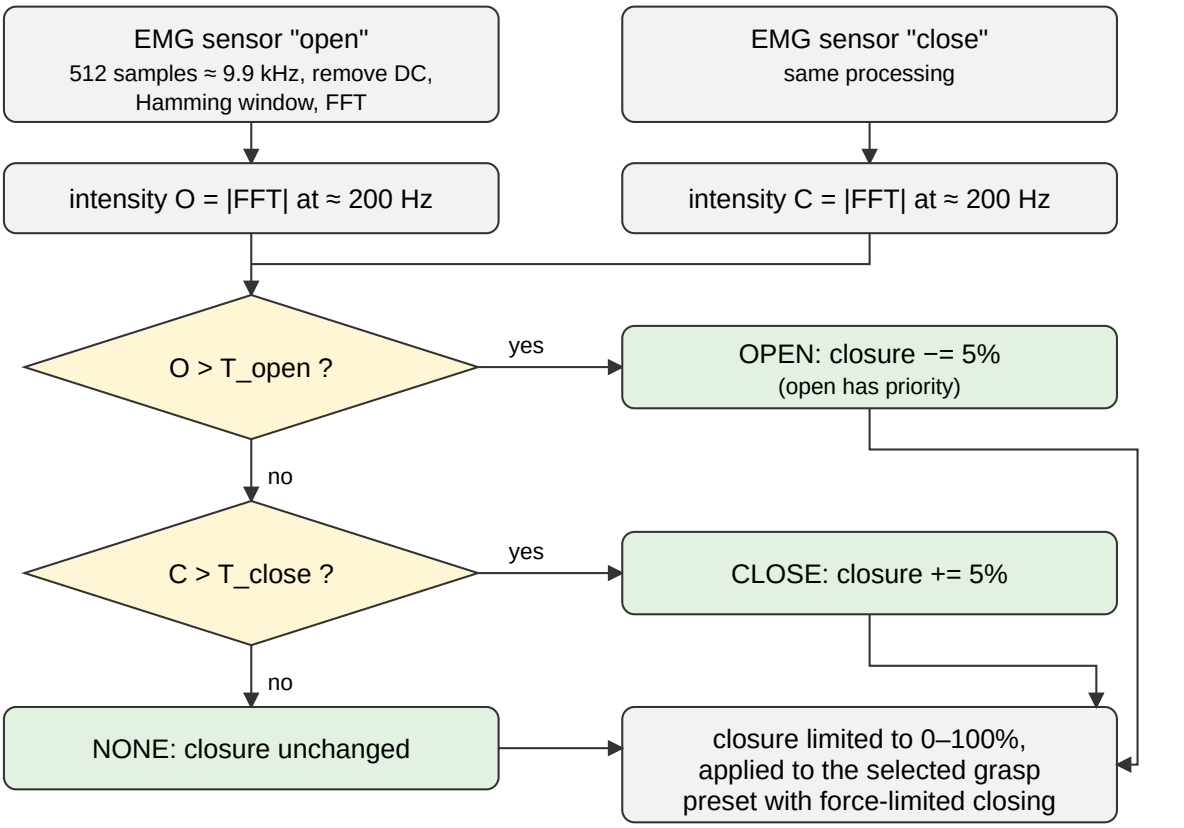


Fig. 8 — Flow chart of the two-channel EMG decision rule (prosthetic firmware).

Alternative embodiments

Tubes may be PTFE, nylon, polyethylene or coil-wound steel housings, one per tendon end or multi-lumen, separate or formed as channels inside the printed skeleton, and may extend outside the hand to actuators in the forearm, on the robot or remote. The two-part spool may have equal or different diameters, stepped, conical or cam-shaped sections, parts rotatable with respect to each other for pre-tensioning and then locked, more than two parts for coupled fingers, and tendon locking by bolt, set screw, clamp plate or wedge. Upper fingers may use one identical module, or modules of individual lengths and proportions for each position (index, middle, ring, little), scaled or anthropometric sizes, or a common module with position-specific phalanx lengths or covers. Two or more fingers may be coupled directly on one spool, with equal or different diameters per finger to set their relative speed. The actuator platform may carry one or more actuators (thumb rotation, thumb flexion, wrist) and be removable as a module. Feedback may be sent over serial, Bluetooth, Wi-Fi, USB, CAN or Ethernet to ROS, ROS 2 or other middleware, and used for slip detection, thermal protection, health monitoring and learning-based control. EMG control may use N channels and FFT, machine-learning or combined processing. The wrist may rotate about the X, Y and/or Z axis. Further variants are listed in `VARIANTS.md` of the repository, which is part of this disclosure, and in the Rebelia V1 disclosure.

Advantages

- Every tendon end has a fixed, protected path; no tendon barrier or pulleys.
- Tendon path length independent of other moving parts; less friction and wear.
- Two-part spool simplifies threading and tensioning, and allows different take-up and release diameters.
- Two fingers coupled on one actuator with no extra parts.
- Thumb rotation actuator assembled and replaced as a module.
- One identical, interchangeable module for all four upper fingers, supplied with tendons pre-installed and replaceable individually.
- Rich feedback for safe operation and integration with ROS.

Applications

Robot end effectors, collaborative and service robots, humanoid robots, teleoperation, research and education, upper-limb prostheses, rehabilitation and exoskeleton devices.